Qiong Yan; Peide Yang (corresponding author)
Fujian Jiangxia University, Fuzhou 350108, China; Xiamen University Tan Kah Kee College, Xiamen, China
Emails: 381007382@qq.com; yangpd@xujc.com

08 November 2026

To the Editor-in-Chief, Solar Energy (Elsevier)

Dear Editor,

Please consider the enclosed original research article "A differentiable, Poynting-consistent transfer-matrix and drift–diffusion kernel for co-design of thin-film solar cells", by Qiong Yan and Peide Yang, for publication in Solar Energy. It is original work, has not been published previously and is not under consideration elsewhere.

Solar Energy regularly publishes transfer-matrix optical design of thin-film cells and drift-diffusion/SCAPS device optimisation. Our contribution joins these two lines of work: a fully coherent, differentiable transfer-matrix solver is coupled directly to a self-consistent one-dimensional drift-diffusion solver, without a surrogate model and without interpolating the optical field onto the electrical mesh. Existing differentiable photovoltaic programs use Beer-Lambert absorption and therefore cannot represent standing-wave interference.

The generation rate is Poynting-consistent; the optical model satisfies the absolute energy balance to 1.44e-6 and the Beer-Lambert limit to 1.57e-8. Implicit differentiation of the converged electrical state costs one transposed Jacobian solve, 1.10 times a forward pass, and agrees with fourth-order Richardson finite differences across the optical, electrical and composite paths. Three-parameter inverse design recovers all parameters to within 0.002% in 44 evaluations.

We would draw attention to the reporting of limitations. The derivative-free benchmark uses true function-call counts; the pre-registered tenfold reduction in eight dimensions is NOT met (about threefold at moderate accuracy, neither optimiser reaching 1e-4). The production mesh carries a 1.52% discretisation uncertainty. All data are synthetic: no experimental calibration, no SCAPS-1D cross-validation and no efficiency record are claimed. We regard this explicit accounting, together with the accompanying reproducible suite, as part of the methodological contribution.

A one-factor 8-D/16-D control (E4-C) and a noisy multi-start comparison against five derivative-free baselines (E5) are reported as independent supplements. They are shipped with their recorded histories and do not revise the negative E4 verdict.

Both authors approved the submission. The work is original, has no competing interests and received no specific funding (to be confirmed by the authors). Generative AI was used only for language editing and manuscript source preparation, as disclosed in the manuscript. The deterministic experiment suite, result files, figure scripts and consistency checker are publicly available from https://github.com/qiantanzhijiao/TMM-DD-Codesign at the time of submission (the same link appears in the manuscript); a frozen Zenodo archive will be linked before publication. We can suggest reviewers on request.

Enclosures: manuscript (LaTeX, PDF, Word), 5 figures, 5 tables, Highlights and the declarations in the manuscript.

Yours sincerely,

Peide Yang (corresponding author)
Xiamen University Tan Kah Kee College; yangpd@xujc.com